MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER
AND-OR-INVERT GATES, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-0038510/41B(USAF), dated 1 May 1974.

Pages 3 and 4

TABLE I, device type 01: Delete and substitute new table I as printed on pages 2 and 3 of this amendment.

Pages 5 and 6

TABLE I, device type 04: Delete and substitute new table I as printed on pages 4 and 5 of this amendment.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-47)

FSC 5962

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TABLE I. (Device type 01) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
1*	V_{OL}	2.0 V	GND	GND	4.5 V	GND						GND	2.0 mA	2.0 V	2.0 V		0.3	±0.03	V
2*		GND	2.0 V	2.0 V		2.0 V							2.0 mA	GND	GND				
3*							2.0 V	2.0 V	GND	GND	2.0 mA								
4*	V_{OL}						GND	GND	2.0 V	2.0 V	2.0 mA						0.3	±0.03	
5*	V_{OH}	0.7 V	0.7 V	5.5 V		5.5 V							-100 µA	5.5 V	5.5 V	2.4		±0.12	
6*			5.5 V	0.7 V		5.5 V							-100 µA	5.5 V	0.7 V				
7*							0.7 V	5.5 V	5.5 V	0.7 V	-100 µA								
8*	V_{OH}				4.5 V		5.5 V	0.7 V	0.7 V	5.5 V	-100 µA					2.4		±0.12	V
9*	V_{OH}	5.5 V	5.5 V	5.5 V	4.5 V	0.7 V							-100 µA	0.7 V	5.5 V	2.4		±0.12	V
10	I_{OS}	GND	GND	GND	5.5 V	GND							GND	GND	GND	-3	-15		mA
11	I_{OS}						GND	GND	GND	GND	GND					-3	-15		mA
12*	I_{IH1}	2.4 V	GND	GND		GND								GND	GND		10	10X or 1 µA	µA
13*		GND	2.4 V	GND		GND													
14*			GND	2.4 V		GND													
15*				GND		2.4 V	GND												
16*						GND	2.4 V	GND											
17*							GND	2.4 V	GND										
18*								GND	2.4 V	GND				GND					
19*									GND	2.4 V				GND					
20*										GND				2.4 V	GND				
21*	I_{IH1}	GND												GND	2.4 V		10	10X or 1 µA	
22	I_{IH2}	5.5 V	GND												GND		100		
23		GND	5.5 V	GND															
24			GND	5.5 V		GND													
25				GND		5.5 V	GND												
26						GND	5.5 V	GND											
27							GND	5.5 V	GND										
28								GND	5.5 V	GND									
29								GND	GND	5.5 V				GND					
30	I_{IH2}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND		GND		5.5 V	GND		100		µA

See notes at end of table.

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TABLE I. (Device type 01) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
31	I_{IH2}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND		GND		GND	5.5 V		100		µA
32	I_{IL}	0.3 V	5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				5.5 V			-.18		mA
33		5.5 V	0.3 V	5.5 V		5.5 V													
34			5.5 V	0.3 V		5.5 V													
5				5.5 V		0.3 V	5.5 V												
36						5.5 V	0.3 V	5.5 V											
37							5.5 V	0.3 V	5.5 V										
38								5.5 V	0.3 V	5.5 V									
39									5.5 V	0.3 V				5.5 V					
40									5.5 V	5.5 V				0.3 V	5.5 V				
41	I_{IL}	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				5.5 V	0.3 V		-.18		
42	I_{CCL}	5.0 V	5.0 V	5.0 V		5.0 V	5.0 V	5.0 V	5.0 V	5.0 V				5.0 V	5.0 V		1.3		
43	I_{CCH}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND		GND		GND	GND		0.8		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE 1. (Device type 04) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
1*	V_{OL}	2.0 V	GND	GND	4.5 V	GND	GND	GND	GND	GND	2.0 mA	GND		2.0 V	2.0 V		0.3	±0.03	V
2*	↓	GND	2.0 V	2.0 V	↓	GND	GND	GND	GND	GND	↓	↓		GND	GND		↓	↓	↓
3*	↓	GND	GND	GND	↓	2.0 V	2.0 V	GND	GND	GND	↓	↓		GND	GND		↓	↓	↓
4*	V_{OL}	GND	GND	GND	↓	GND	GND	2.0 V	2.0 V	2.0 V	2.0 mA	↓		GND	GND		0.3	±0.03	↓
5*	V_{OH}	0.7 V	0.7 V	5.5 V	↓	0.7 V	5.5 V	5.5 V	0.7 V	5.5 V	-100 µA	↓		5.5 V	5.5 V	2.4	↓	±0.12	↓
6*	↓	5.5 V	0.7 V	5.5 V	4.5 V	0.7 V	5.5 V	0.7 V	5.5 V	5.5 V	↓	↓		0.7 V	5.5 V	↓	↓	↓	↓
7*	↓	5.5 V	5.5 V	0.7 V	4.5 V	5.5 V	0.7 V	5.5 V	5.5 V	0.7 V	↓	↓		5.5 V	0.7 V	↓	↓	↓	↓
8	I_{OS}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND	GND	↓	↓	GND	GND	-3	-15	↓	mA
9*	I_{IH1}	2.4 V	GND	GND	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	10	10X or 1 µA	µA
10*	↓	GND	2.4 V	GND	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
11*	↓	↓	GND	2.4 V	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
12*	↓	↓	↓	GND	↓	2.4 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
13*	↓	↓	↓	↓	↓	GND	2.4 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
14*	↓	↓	↓	↓	↓	↓	GND	2.4 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
15*	↓	↓	↓	↓	↓	↓	↓	GND	2.4 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓
16*	↓	↓	↓	↓	↓	↓	↓	↓	GND	2.4 V	↓	↓	↓	GND	↓	↓	↓	↓	↓
17*	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	↓	↓	2.4 V	GND	↓	↓	↓	↓
18*	I_{IH1}	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	2.4 V	↓	10	10X or 1 µA	↓
19	I_{IH2}	5.5 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	GND	↓	100	↓	↓
20	↓	GND	5.5 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
21	↓	↓	GND	5.5 V	↓	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
22	↓	↓	↓	GND	↓	5.5 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
23	↓	↓	↓	↓	↓	GND	5.5 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
24	↓	↓	↓	↓	↓	↓	GND	5.5 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
25	↓	↓	↓	↓	↓	↓	↓	GND	5.5 V	GND	↓	↓	↓	↓	↓	↓	↓	↓	↓
26	↓	↓	↓	↓	↓	↓	↓	↓	GND	5.5 V	↓	↓	↓	GND	↓	↓	↓	↓	↓
27	↓	↓	↓	↓	↓	↓	↓	↓	GND	GND	↓	↓	↓	5.5 V	GND	↓	↓	↓	↓
28	I_{IH2}	GND	GND	GND	↓	GND	GND	GND	GND	GND	↓	↓	↓	GND	5.5 V	↓	100	↓	µA
29	I_{IL}	0.3 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	↓	GND	↓	5.5 V	5.5 V	↓	-.18	↓	mA

See notes at end of table.

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TABLE I. (Device type 04) Electrical characteristics tests.

1/2/ TEST NO.	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
30	I_{IL}	5.5 V	0.3 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		GND		5.5 V	5.5 V		-.18		mA
31			5.5 V	0.3 V		5.5 V	5.5 V												
32				5.5 V		0.3 V	5.5 V												
33						5.5 V	0.3 V	5.5 V											
34							5.5 V	0.3 V	5.5 V										
35								5.5 V	0.3 V	5.5 V									
36									5.5 V	0.3 V				5.5 V					
37									5.5 V	5.5 V				0.3 V	5.5 V				
38	I_{IL}	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				5.5 V	0.3 V		-.18		
39	I_{CCL}	5.0 V	5.0 V	5.0 V		5.0 V	5.0 V	5.0 V	5.0 V	5.0 V				5.0 V	5.0 V		0.99		
40	I_{CCH}	GND	GND	GND	5.5 V	GND	GND	GND	GND	GND		GND		GND	GND		0.8		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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